ACCESCO LIVING PRIVATE LIMITED

CIN: U47912WB2025PTC285309

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DPIIT Startup India Certified

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Pull Theory

A demand-led replenishment system for the Accesco shared store. Working draft for discussion, 30 October 2026. Not approved for external use.

A store should hold tomorrow only what the neighbourhood proved it eats today, corrected for the day of the week, and nothing more, because the kitchen absorbs the difference.

1. Why Pull

Quick commerce today is supply-led. Distributors and brands push stock into dark stores weeks ahead, the app shows whatever is on the shelf, and advertising nudges households to buy more of it. The shelf decides what the neighbourhood eats. The cost of being wrong is carried as write-offs, which is why these stores stay lean and show items as unavailable.

Pull reverses the direction. What the neighbourhood actually eats decides what is stocked, what is cooked and what the farms send. Demand flows backwards to the farm every night. The same idea let Toyota outbuild every car company in the world; it has never been applied to fresh commerce in India because no store had a way to absorb the daily gap between what it bought and what it sold.

Accesco has that absorber. The Swadishtt kitchen sits on the same shelf as Grokly. Whatever is bought and not sold is cooked, and whatever sells faster than expected is covered by the kitchen shifting its menu. Pull is therefore safe for Accesco in a way it is not for any dark store.

The household does nothing new. It orders what it wants to eat, in any of the three ways (the ingredients, the dish cooked in our kitchen, or an Accesco cook at the door). The planning happens behind the counter.

2. The replenishment formula

For each item i, the quantity ordered tonight for delivery tomorrow morning:

Argha's Formula

$$Q(i, t+1) = \max(0 , D(i, t+1) \times (1 + b_i) - OnHand(i, t))$$

where the demand estimate for tomorrow is:

Argha's Formula

$$D(i, t+1) = A(i, w) \times F(t+1) \times M(t+1) \times R(i)$$

Term	Meaning	How it is set
A(i, w)	Trailing 4-week average of units sold on the same weekday w, with dishes converted into ingredients through their recipes	From Accesco sales. Day one uses the seed list (section 7).
F(t+1)	Festival or fasting factor	1.00 on ordinary days; set per event from the local calendar (Ugadi, Ganesha festival, Deepavali, Ramzan, Ayudha Puja).
M(t+1)	Month-cycle factor (salary-day effect)	Above 1 in the first week of the month, below 1 in the last week (section 4).

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Term	Meaning	How it is set
R(i)	Rejection factor from Doorstep Check	1.00 normally. Falls when an item is rejected often; the response is to change the farm or lot, not to order more.
b_i	Buffer by perishability	Small, because the kitchen absorbs overflow (section 3).
OnHand(i, t)	Sound stock on the shelf at closing	Counted at 9 pm; stock already routed to the kitchen is excluded.

Using a trailing weekday average rather than yesterday alone is what makes Pull stable. One rainy Tuesday no longer distorts Wednesday, and the average converges to the neighbourhood's real rhythm within four weeks.

2a. Mathematical derivation of Argha's Formula

Step 1. The order-up-to level. Pull is a base-stock policy: every night the store brings each item up to a target level S(i, t+1) and never above it. The order is the gap between the target and what is already on the shelf, and it cannot be negative:

Argha's Formula

$$Q(i, t+1) = \max(0 , S(i, t+1) - OnHand(i, t))$$

Step 2. The target is expected demand plus a buffer. Let D(i, t+1) be the expected demand for tomorrow and b_i a buffer fraction, so that S = D x (1 + b_i). Substituting gives the main formula.

Step 3. Expected demand from the weekday average. Let x(i, w, j) be the units sold on the j-th most recent occurrence of weekday w (dishes converted to ingredients through their recipes). With k = 4 weeks of history:

Argha's Formula

$$A(i, w) = (1 / k) \times \text{SUM over } j = 1..k \text{ of } x(i, w, j)$$

The weekday average already contains the day-of-week effect, because Saturdays are only ever compared with Saturdays. The explicit day constants in section 4 are used only while k is small (the first weeks), when A is estimated from the overall daily mean multiplied by the day constant: A(i, w) = mean(i) x c(w).

Step 4. Multiplicative corrections. Known one-off effects enter as factors, not additions, because they scale demand proportionally: a festival that lifts dinner orders by 20% lifts a large store and a small store by the same proportion.

Argha's Formula

$$D(i, t+1) = A(i, w) \times F(t+1) \times M(t+1) \times R(i)$$

Step 5. Why the buffer can be small: the newsvendor view. For one item, let Cu be the cost of being short by one unit (lost margin plus a disappointed household) and Co the cost of having one unit too many. Classical inventory theory sets the target so that the probability of meeting demand equals the critical ratio:

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Argha's Formula

P(demand <= S) = Cu / (Cu + Co)

In a dark store, Co is the full cost of the unit, because surplus fresh stock is written off. In the Accesco store, Co = c - v, where c is the purchase cost and v is the value the kitchen recovers by cooking the surplus. For cookable items v is close to c, so Co is small, the critical ratio is high, and the store can afford to target a high fill rate with a modest buffer. The buffer b_i is the fraction that lifts P(demand <= S) to the chosen fill rate (97% in section 8) for that item's demand spread, capped by the next step.

Step 6. The kitchen capacity constraint. Expected overflow for an item is E[max(0, S - demand)]. Summed across fresh items, it must stay within the kitchen's daily absorption capacity K:

Argha's Formula

SUM over fresh items i of E[max(0, S(i) - demand(i))] <= K

If the sum exceeds K, buffers are reduced from the most perishable items first until the constraint holds. This is the only place Pull trades fill rate for waste, and it does so by rule, not by judgement.

Step 7. Convergence. As weeks of sales accumulate, A(i, w) converges to the neighbourhood's true weekday demand and the starting priors drop out. The formula is therefore self-correcting: every day of sales is a new observation that tightens tomorrow's order.

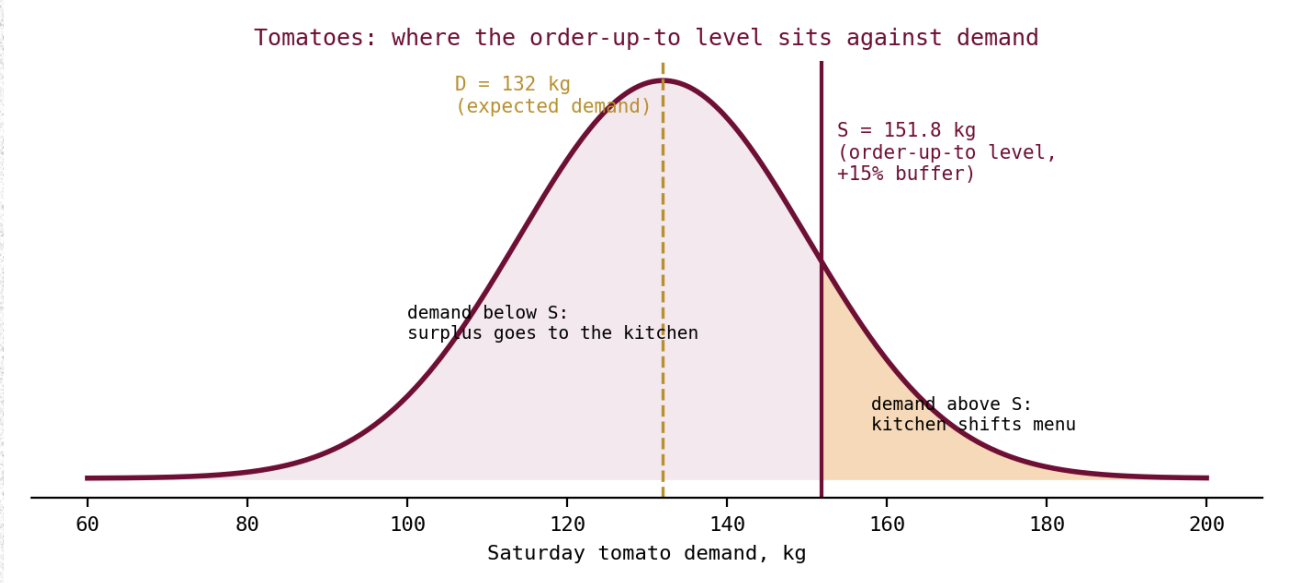

Figure 1. The order-up-to level S against the spread of Saturday tomato demand (worked example A). Demand below S leaves a surplus that the kitchen cooks; demand above S is covered by the kitchen shifting its menu. In a dark store the left region is a write-off; here it is dinner.

3. Buffers by category (starting priors)

Category	Buffer b_i	Reasoning
Leafy greens, herbs	10%	Shortest life; overflow is cooked the same day, so a small buffer is enough.
Other fresh produce	15%	Two to four days of life; the kitchen absorbs overflow within that window.

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Category	Buffer b_i	Reasoning
Dairy, bread, paneer	10%	Daily delivery; shortfall is costly, overflow is cooked.
Packaged goods (FMCG)	25%	Long shelf life; a stock-out costs more than overstock, and nothing is wasted.

Hard constraint: the expected overflow across all fresh items for a day must not exceed the kitchen's absorption capacity for that day (what it can cook into meals and sell). If it does, buffers are reduced proportionally, starting with the most perishable items.

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4. Day-of-week and month-cycle constants

Important: web research was unavailable when this draft was written, so the constants below are starting priors with their reasoning, not published data. They are to be run for two weeks and then replaced by Rajajinagar's own sales, and cross-checked against the public sources listed in section 5.

Day	Groceries	Meals and cooks	Reasoning behind the prior
Monday	0.95	0.90	Fridges are full from the weekend; eating out resets to cooking at home.
Tuesday	0.95	0.90	Lowest-effort weekday; routine meals.
Wednesday	0.97	0.95	Mid-week top-up of fresh items.
Thursday	0.98	1.00	Pre-weekend top-up begins; meal orders rise.
Friday	1.03	1.15	Weekend eating starts Friday night; cook bookings rise.
Saturday	1.10	1.20	Weekly grocery stock-up and the peak dinner night.
Sunday	1.02	1.10	Family meals and Sunday cooking; groceries dip after Saturday's stock-up.

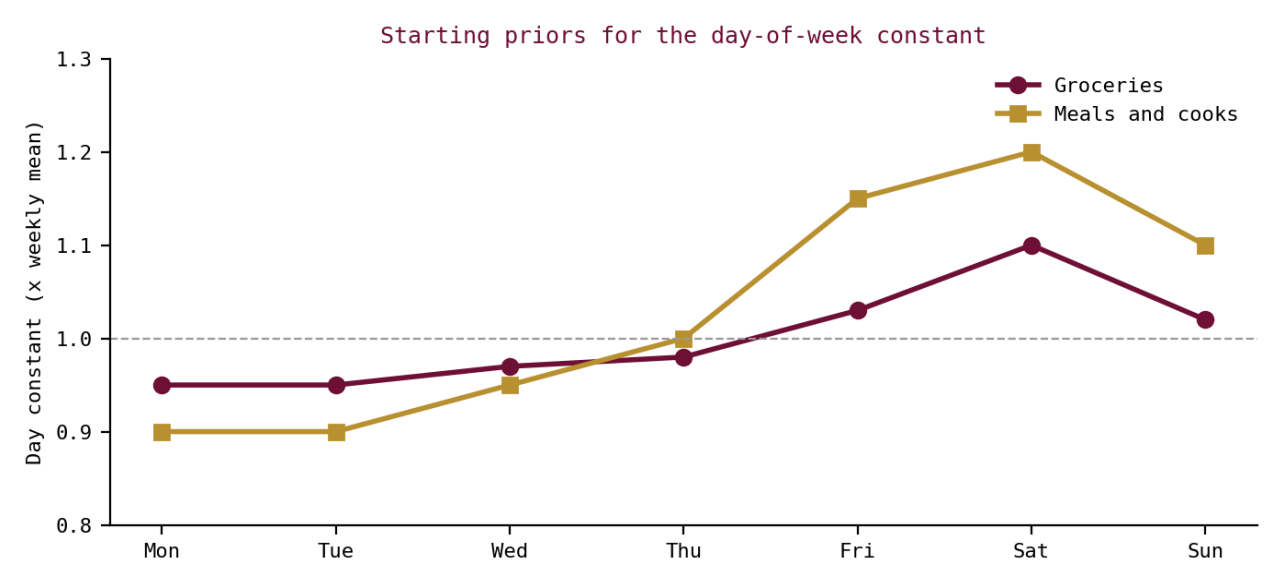

Figure 2. The starting priors for the day-of-week constant, groceries and meals, indexed to the weekly mean (1.00). They are replaced by Rajajinagar's own sales after two weeks.

Shape of the table: it reflects the widely reported pattern that food delivery peaks on Friday and Saturday evenings and dips on Monday and Tuesday, and that grocery baskets are largest on Saturdays. These patterns could not be verified or cited at the time of writing.

Month-cycle factor M, starting prior:

Days of the month	M	Reasoning
1 to 7	1.08	Salaried households restock after payday.
8 to 22	1.00	Baseline.
23 to 31	0.92	Budgets tighten before the next salary; baskets shrink.

Public household expenditure surveys do not report spending by day of the month, so M must be derived from Accesco's own first month of sales.

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5. How to derive the real constants (sources to use once web access is on)

Source	What it gives	How it becomes a constant
Google Trends (Bengaluru, past 12 months): 'order food online', 'groceries online', 'Swiggy', 'Blinkit', 'Zepto'	Relative weekly search curves	Average each weekday's index; divide by the weekly mean. Saturday at 15.7% of the week against a flat 14.3% gives 1.10.
Eternal (Zomato) and Swiggy investor presentations and annual reports	Commentary on weekend order share and dinner peaks	Weekend share versus weekday share sets the Friday to Sunday multipliers for meals.
Redseer, NRAI (India Food Services Report) and Bain quick-commerce reports	Weekday versus weekend demand mix	Same method; use for groceries and meals separately.
Statista: day-of-week food-delivery distributions, India	Published distributions where available	Index to the weekly mean.
Accesco survey (n=701)	63% buy online at least two to three times a week	Sets baseline frequency, not the weekday shape.
Accesco sales (after two weeks)	Actual units by weekday and item	Replaces every prior above; the trailing average takes over from week four.

Rule for citing: each constant in production carries the source and the date it was derived, and is recomputed monthly.

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6. Worked examples

Example A: tomatoes, order placed Friday night for Saturday.

Step	Value
Trailing Saturday average A	120 kg
Saturday grocery constant	1.10
Festival F, month M, rejection R	1.00, 1.00, 1.00
Demand estimate D	120 x 1.10 = 132 kg
Buffer b (other produce) 15%	132 x 1.15 = 151.8 kg
On hand at 9 pm Friday	20 kg
Order to FPO by 9:30 pm	151.8 - 20 = 131.8, rounded to 132 kg

If Saturday sells 110 kg, about 22 kg moves to the kitchen's Sunday menu (a tomato-based gravy day). If Saturday sells 150 kg, the kitchen shifts its menu away from tomato-heavy dishes and the shelf shows true availability. Either way the error becomes a meal, not a write-off.

Example B: paneer, order placed Monday night for Tuesday.

Step	Value
Trailing Tuesday average A (shelf sales plus kitchen use)	18 kg
Tuesday grocery constant	0.95
Demand estimate D	17.1 kg
Buffer b (dairy) 10%	18.8 kg
On hand at 9 pm Monday	4 kg
Order to the dairy supplier	14.8, rounded to 15 kg

Example C: a cook booked at 5 pm for 7 pm. The booking's dish list is added to the day's demand for the kitchen prep plan and to tonight's A(i, w) for tomorrow's order, so a rising trend in cook bookings pulls more of those ingredients onto the shelf the next morning.

7. The daily cycle and the first two weeks

Time	What happens
9:00 pm	Store closes. Sales, dishes, cook bookings and doorstep rejections are totalled; dishes are converted into ingredients.
9:30 pm	Replenishment is computed and sent: fresh to FPOs (daily pull), packaged goods to brands and distributors (1 to 2 day pull against minimum and maximum levels).
4:00 am	FPO trucks and distributor vans deliver. Stock is counted in; lot numbers recorded.
5 am to 9 pm	The shelf sells, the kitchen cooks, cooks go out from 6 am. Fresh Routing moves stock nearing its limit into the kitchen during the day.
Continuous	Doorstep Check results flow back to R(i); kitchen absorption is logged.

Bootstrapping: day one has no yesterday. Seed the shelf from the ICP study and survey (the top 100 items and 20 dishes for Rajajinagar), run Pull from day two, and let the weekday constants settle over two weeks. The first week will be wrong; the kitchen pays for the error, which is its job.

8. What to measure weekly

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Metric	Target	Why
Fill rate (item in stock when ordered)	above 97%	Pull must not create stock-outs.
Waste (neither sold nor cooked)	under 2% of stock	The kitchen should absorb nearly all overflow.
Kitchen absorption	above 80% of overflow	Measures the absorber.
Forecast error by weekday	falling week on week	Shows the constants converging.
FPO fill against the nightly order	above 90%	Daily pull depends on farms delivering.
Hours from harvest to door (fresh)	under 24	The freshness Pull is meant to deliver.

9. Risks and how Pull handles them

Risk	Response
A farm misses the nightly order	Backup supplier per item; the kitchen substitutes within dishes; the shelf shows true stock.
Demand spikes (festival, rain, a viral dish)	F and weather adjustments; instant delivery remains; the kitchen cooks from what is on the shelf.
Kitchen capacity is exceeded by overflow	Buffers are cut automatically from the most perishable items first.
Constants are wrong at launch	They are priors by design; two weeks of sales replace them.
A new store has no history	Seed from the nearest store's weekday shape, then let its own data take over.

Summary: Pull makes Accesco demand-led with a single nightly calculation, adds no step for the household, and is only possible because the kitchen and the shelf are one inventory. Supply-led stores cannot run it; their overflow has nowhere to go.